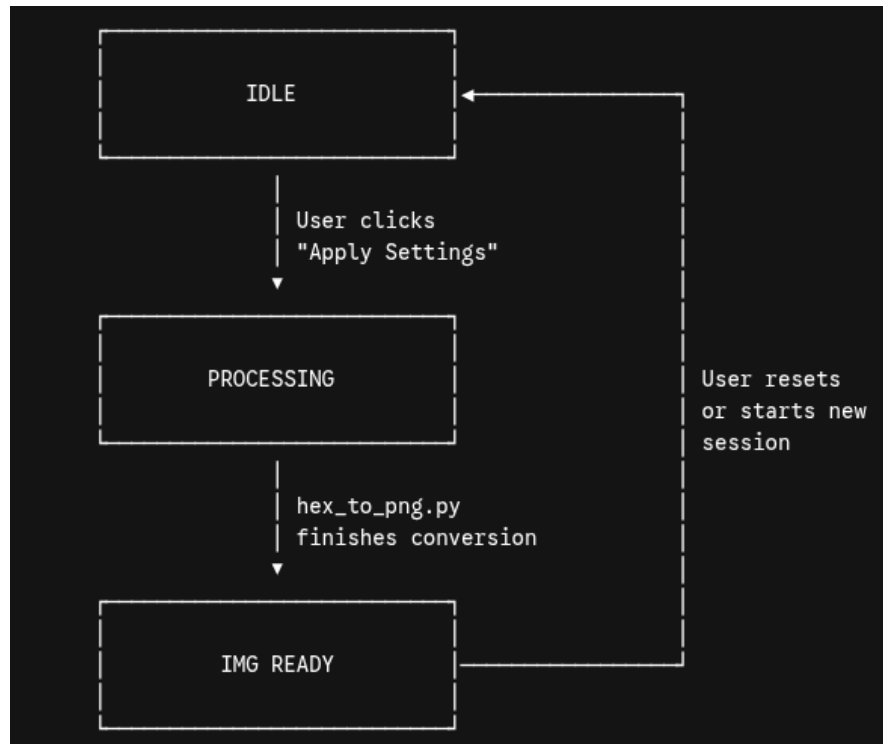


1. GUI State Machine Diagram

This diagram focuses strictly on the states of your GUI (`IDLE`, `Processing`, `img Ready`) and the triggers that cause transitions.



2. System Flowchart (Activity Diagram)

This diagram maps out the end-to-end logic, including script executions, file creations, conditional checks (`debugging` and `input source`), and simulation handling.

Breakdown of the Technical Pipeline:

- **The Bridge (`pipeline_config.txt`):** Acts as the communication link between Python GUI and Python runner (`runProject.py`) to dictate the environment layout (GUI vs. CLI).
- **The Hardware Config (`configuration.vh`):** Acts as the compile-time parameter file for Verilog Testbench (`tb.v`), switching the multiplexing logic between Test Pattern Generation (TPG) and raw image streaming.

